Chapter 3

(Absolutely) Continuous Random Variables

3.1 Example

Example 3.1. Let X be a random variable representing a number chosen uniformly at random from the interval $[0,1]$. Observe that $\mathrm{Im}(X) = [0,1]$, which is not a countable set, so X is not a discrete random variable. This means that X *does not have a probability mass function.* However, X *does* have a cumulative distribution function, which we can find as follows.
First of all,

$$\begin{array}{lrcll} \text{for } x < 0, & F_X(x) & = & \mathbb{P}(X \leqslant x) = 0 & \text{(since we can't have } X \leqslant x < 0), \\ \text{for } x \geqslant 1, & F_X(x) & = & \mathbb{P}(X \leqslant x) = 1 & \text{(since we certainly do have } X \leqslant 1 \leqslant x) \end{array}$$

Now recall from Example 1.6 that for a number X chosen uniformly at random from the interval $[0,1]$, we have $\mathbb{P}(X \in [a,b]) = b - a$ for any a, b with $0 \leqslant a \leqslant b \leqslant 1$. So,

$$\text{for } 0 \leqslant x < 1, \qquad F_X(x) \;=\; \mathbb{P}(X \leqslant x) = \mathbb{P}(X \in [0,x]) = x - 0 = x$$

Thus, the cumulative distribution function of X is

$$F_X(x) = \begin{cases} 0 & x < 0 \\ x & 0 \leq x < 1 \\ 1 & x \geq 1 \end{cases}$$

Notice that the distribution function $F_X(x)$ is defined piece-wise.
A sketch of the cdf F_X is given in Figure 3.1.
Notice that $F_X(x)$ is a continuous function, and is differentiable everywhere except at $x = 0$ and $x = 1$.
Observe that for $x \in (0,1)$

$$\mathbb{P}(X = x) = \lim_{n\to\infty} F_X(x) - F_X(x - \frac{1}{n}) = \lim_{n\to\infty} \frac{1}{n} = 0.$$

This means that working with the uniform distribution is fundamentally different from using discrete random variables.

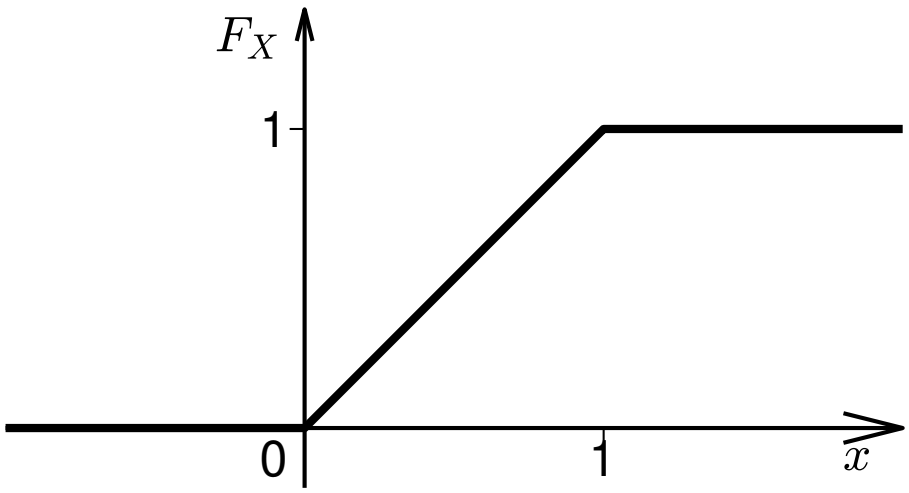


Figure 3.1: Sketch of $F_X(x)$

3.2 Probability density function (pdf)

Although *every* random variable X has a cdf F_X, for discrete random variables it is often easier to work with the probability mass function. Similarly, for (absolutely) continuous random variables it is often easier to work with the probability density function, defined as follows.

Definition: Probability density function
Suppose X is a random variable with distribution function F_X, and suppose that there is a function $f_X : \mathbb{R} \to \mathbb{R}$ s.t. $\forall a < b$,

$$\mathbb{P}(a < X \leqslant b) = F(b) - F(a) = \int_a^b f_X(x)dx \qquad (*)$$

Thenf_X is called the (probability) density function (pdf) of X, and X is said to be an (absolutely) continuous random variable.

Remarks
Suppose that X is a r.v. with probability density function f_X. Then:

1. Letting $a \to -\infty$ in equation $(*)$ above, we obtain

$$\boxed{F(b) = \int_{-\infty}^b f_X(x)dx \qquad \forall b \in \mathbb{R}}$$

   If X has a density function f_X, then we can find the cdf F_X for X by *integrating* f_X

2. Since F_X is obtained by integrating f_X, then conversely, f_X may be obtained by differentiating F_X. That is, the probability density function $f_X(x)$ is given by

$$\boxed{f_X(x) = \frac{dF_X}{dx} \quad \text{for all } x \text{ such that the derivative exists.}}$$

3. Recall that a function $F : [a, b] \to \mathbb{R}$ is absolutely continuous on $[a, b]$ if there exists an integrable function $g : [a, b] \to \mathbb{R}$ such that for all $x \in [a, b]$

   $F(x) = F(a) + \int_a^x g(s)ds.$

Therefore an (absolutely) continuous random variable is a random variable whose cdf is absolutely continuous. There are examples of functions which are continuous but not absolutely continuous, a classic example of this is the Cantor function (see Section 7 of Grimmett, G., & Welsh, D. (1990). Probability: An Introduction for details on the Cantor function).

Example 3.1 continued. Suppose random variable X represents a number picked uniformly at random from the interval $[0, 1]$. The cumulative distribution function of X is

$$F_X(x) = \begin{cases} 0 & x < 0 \\ x & 0 \leqslant x < 1 \\ 1 & x \geqslant 1 \end{cases}$$

To obtain the probability density function $f_X(x)$, differentiate:

$$f_X(x) = \frac{dF_X}{dx} = \begin{cases} 0 & x < 0 \\ 1 & 0 < x < 1 \\ 0 & x < 1 \end{cases}$$

Here is a sketch of the probability density function f_X:

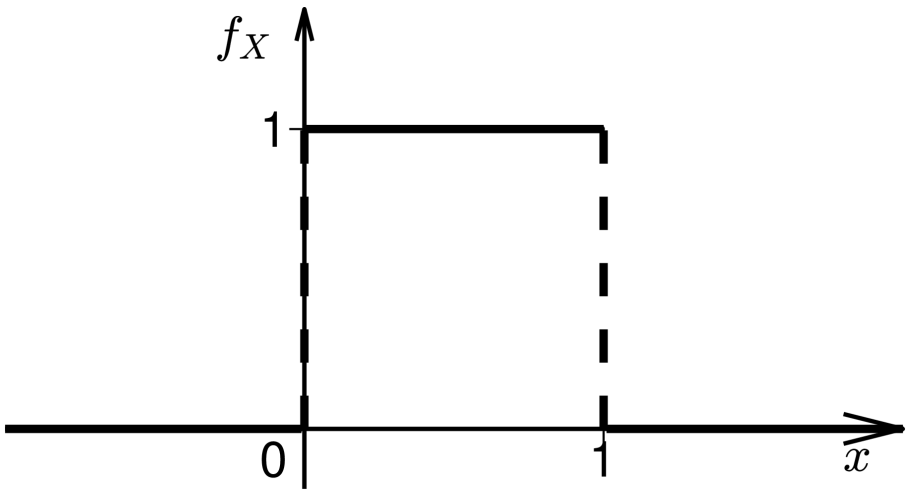


Figure 3.2: Sketch of $f_X(x)$

Notice that for $x \in \mathrm{Im}X$, the value of $f_X(x)$ is constant, corresponding to probability being uniformly distributed across the interval $[0, 1]$.

Remarks
Suppose that X is a random variable with probability density function f_X. Then:

1. From the definition of differentiation, for all x such that the derivative exists,

$$f_X(x) = \frac{dF_X}{dx} = \lim_{\delta x \downarrow 0} \frac{\mathbb{P}(x \leqslant X < x + \delta x)}{\delta x}$$

We can see from this that $f_X(x) \geq 0$, and that the value of $f_X(x)$ indicates, in some sense, how likely the random variable X is to be close to the value x, but that $f_X(x)$ is not a probability. To obtain probabilities, we need to integrate $f_X(x)$.

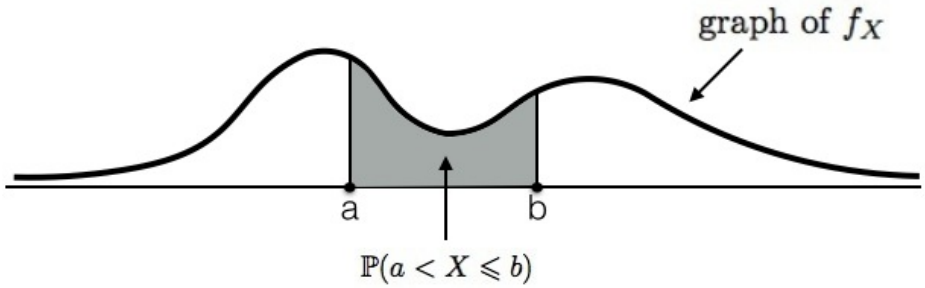
2. $\forall b \in \mathbb{R}$ we have, using continuity of probability measures,

$$\mathbb{P}(X = b) = \lim_{a \uparrow b} \mathbb{P}(a < X \leqslant b) = \lim_{a \uparrow b} \int_a^b f_X(x)\,\mathrm{d}x = 0$$

Since $\mathbb{P}(X = b) = 0$ for all $b \in \mathbb{R}$, this means that the cumulative distribution function F_X does not have jumps. That is, if a random variable X has a density function, then the cumulative distribution function F_X is a *continuous* function. This is why we say that X is an (absolutely) *continuous* random variable when it has a probability density f_X.

3. For any $a < b$,

$$\mathbb{P}(a \leqslant X \leqslant b) = \mathbb{P}(a \leqslant X < b) = \mathbb{P}(a < X < b) = \mathbb{P}(a < X \leqslant b)$$
$$= \int_a^b f_X(x)\,\mathrm{d}x = (\text{area under } f_X \text{ between } a \text{ and } b)$$



4. Since $F(b) = \int_{-\infty}^{b} f_X(x)dx$ and $\lim_{b\to\infty} F(b) = 1$ we get:

$$1 = \lim_{b\to\infty} F(b) = \lim_{b\to\infty} \int_{-\infty}^{b} f_X(x)\,\mathrm{d}x = \int_{-\infty}^{\infty} f_X(x)\,\mathrm{d}x = (\text{total area under graph of } f_X)$$

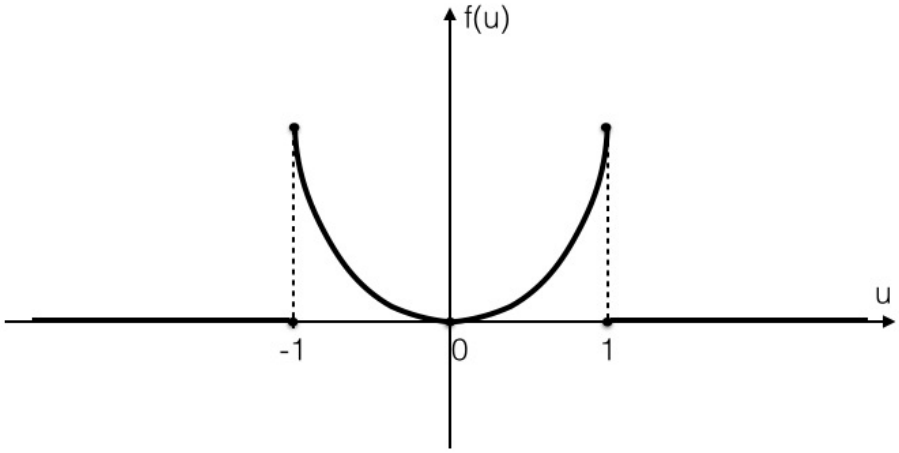
Properties of probability density functions
If X is a r.v. with probability density function f_X, then:

- $f_X(x) \geqslant 0 \quad \forall x \in \mathbb{R}$
- $\int_{-\infty}^{\infty} f_X(x)\,\mathrm{d}x = 1$

Conversely, any function satisfying the above properties is a valid probability density function for some (absolutely) continuous random variable.
(Compare with properties of probability mass functions.)

Example 3.2.

Suppose $f(u) = \begin{cases} \frac{3u^2}{2} & -1 \leqslant u \leqslant 1 \\ 0 & \text{otherwise} \end{cases}$



Is f a density function? If so, what is the corresponding distribution function F?

Answer:

- $f(u) \geqslant 0 \ \ \forall u$, so f is non-negative.
- $\int_{-\infty}^{\infty} f(u)\,\mathrm{d}u = \int_{-1}^{1} \frac{3u^2}{2}\,\mathrm{d}u = \left[\frac{u^3}{2}\right]_{-1}^{1} = 1$ (area under the graph)

therefore f *is* a density function.
If X is a random variable with probability density function f, then $\mathrm{Im}X = [-1, 1]$ because $f(u) > 0$ on the interval $[-1, 1]$, and the cumulative distribution function for X is given by $F_X(x) = \int_{-\infty}^{x} f(u)\,\mathrm{d}u$ for any $x \in \mathbb{R}$.
To determine the cumulative distribution function F_X, we have to consider 3 cases:

$$x < -1: \quad F_X(x) = \int_{-\infty}^{x} f(u)\,\mathrm{d}u = \int_{-\infty}^{x} 0 \ \,\mathrm{d}u = 0$$

$$-1 \leqslant x < 1: \quad F_X(x) = \int_{-\infty}^{x} f(u)\,\mathrm{d}u = \int_{-1}^{x} \frac{3u^2}{2}\,\mathrm{d}u = \frac{x^3+1}{2}$$

$$1 \leqslant x: \quad F_X(x) = \int_{-\infty}^{x} f(u)\,\mathrm{d}u = \int_{-1}^{1} \frac{3u^2}{2}\,\mathrm{d}u = 1$$

So we have

$$F_X(x) = \begin{cases} 0 & x < -1 \\ \frac{x^3+1}{2} & -1 \leqslant x < 1 \\ 1 & 1 \leqslant x \end{cases}$$

Remarks:

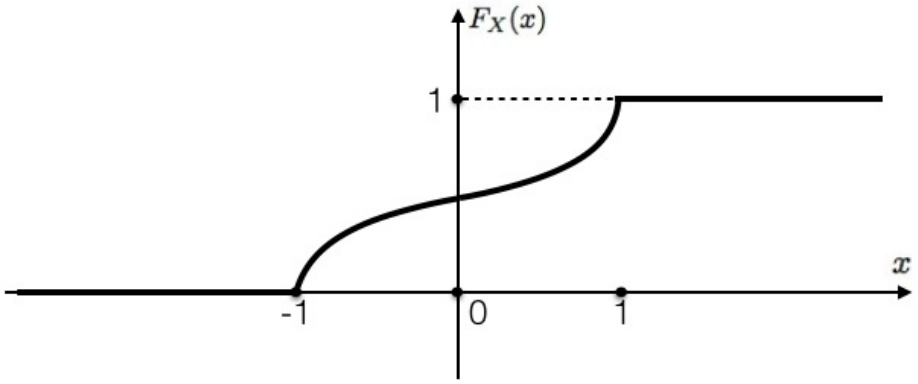


Figure 3.3: Sketch of $F_X(x)$

1. By sketching the graph of the cumulative distribution function F_X we see that it is continuous although the density $f(u)$ is not a continuous function. In general, a probability density function does not have to be continuous. Not all cumulative distribution functions correspond to a probability density function (e.g. discrete random variables don't have a probability density function), but any cumulative distribution function that corresponds to a probability density function is a continuous function.
2. The graph of a density f_X for a random variable X is often easier to interpret than the graph of its cumulative distribution function F_X.

Example 3.2 continued. Suppose X is a random variable with probability density function

$$f_X(u) = \begin{cases} \frac{3u^2}{2} & -1 \leqslant u \leqslant 1 \\ 0 & \text{otherwise} \end{cases}$$

then we can *see* from the graph of f_X that (for example)

$$\mathbb{P}(|X| \geqslant 1/2) > \mathbb{P}(|X| < 1/2)$$

by looking at the *area* under the graph of f_X.

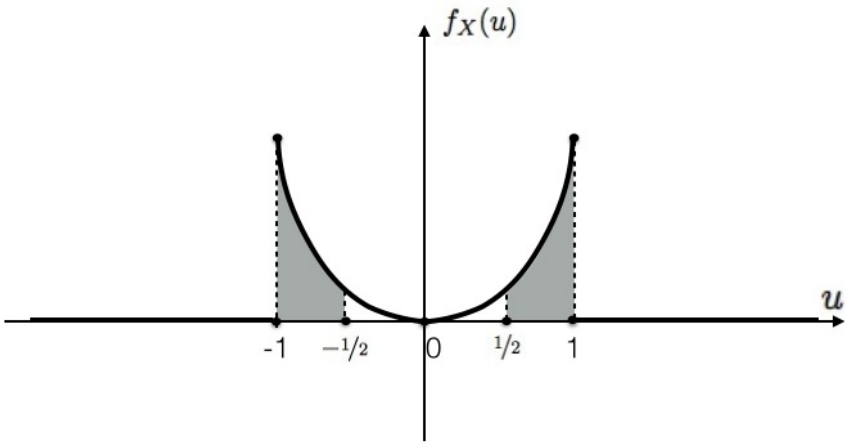


Figure 3.4: Area of shaded region $= \mathbb{P}(|X| \geqslant \frac{1}{2})$

Key point: It can be useful to sketch the probability density function f_X for the r.v. X in order to get an impression about which values X is more or less likely to take.

Question: Suppose X is a random variable with distribution function F_X. How can we tell whether X has a probability density function f_X, and if it does have a probability density function f_X, how can we find f_X?

Answer: If F_X is continuous and differentiable 'almost everywhere' then X has a probability density function which can be found by differentiating F_X

Remark 3.1 (Beyond the course material). For our purposes if F_Y is differentiable at all but a countable set of points then it is differentiable 'almost everywhere'. Although the meaning of 'almost everywhere' is more general than this and goes beyond what we need in this course. A function has a given property 'almost everywhere' is the set of points S where the property fails is sufficiently small. By sufficiently small we mean that if we choose any interval $[a, b]$ with $a < b$ and draw a point U uniformly at random on that interval then the probability $U \in S$ is zero.

Example 3.3. Suppose Y is a random variable with cumulative distribution function F_Y given by

$$F_Y(y) = \begin{cases} 0 & y < 0 \\ \frac{y^2}{2} & 0 \leqslant y \leqslant 1 \\ 1 - \frac{(2-y)^2}{2} & 1 < y \leqslant 2 \\ 1 & 2 < y \end{cases}$$

Is F_Y *continuous*?
To check this, we note that there are 4 'pieces' in the definition of F_Y.
Each piece is continuous, so it is enough to check that the pieces join up (i.e. take the same values) at their boundary points $y = 0$, $y = 1$, and $y = 2$.

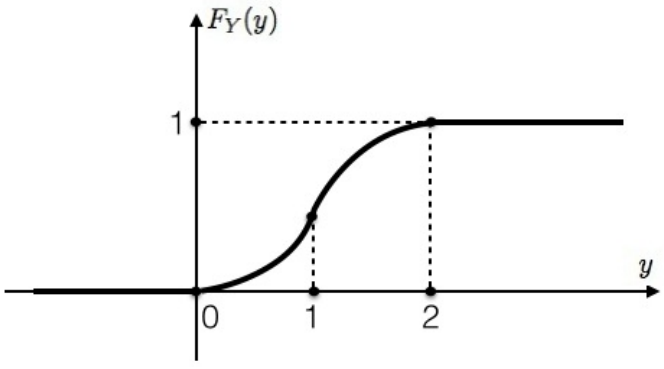


Figure 3.5: Sketch of $F_Y(y)$

From the sketch, we see that F_Y *is* continuous.

Is F_Y differentiable 'almost everywhere'?

Again, since there are 4 pieces in the definition of F_Y and each piece is differentiable, F_Y is differentiable 'almost everywhere' (we say 'almost everywhere' because sometimes there can be

a problem with differentiating at the boundary points).

Therefore Y has a density function $f_Y(y)$ which is given by:

$$f_Y(y) = \begin{cases} 0 & y < 0 \\ y & 0 < y < 1 \\ 2 - y & 1 < y < 2 \\ 0 & 2 < y \end{cases}$$

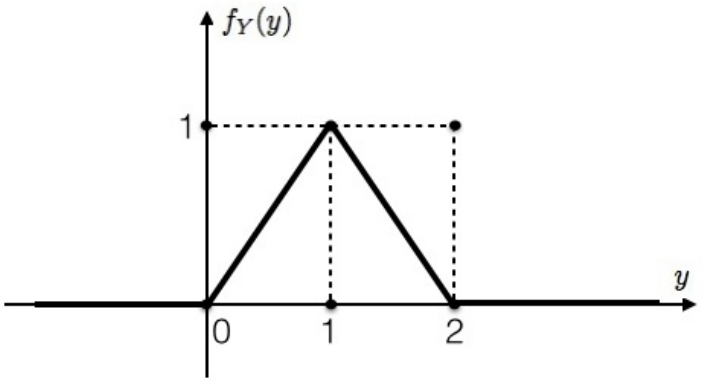


Figure 3.6: Sketch of $f_Y(y)$

3.3 Common examples of continuous random variables

3.3.1 Uniform distribution

Uniform random variables are used to model picking a point at random from the interval $[a, b]$ such that each point in the interval is 'equally likely' to be chosen. If X is a uniform random variable on the interval $[a, b]$, the value taken by X represents the randomly chosen point in the interval $[a, b]$ and $\text{Im}X = [a, b]$.

If X is uniform on $[a, b]$, we write $X \sim \text{U}(a, b)$.

Since X is equally likely to take any value between a and b, we have that for $a \leqslant c \leqslant d \leqslant b$:

$$\mathbb{P}(c \leqslant X \leqslant d) = \frac{\text{length of } [c, d]}{\text{length of } [a, b]} = \frac{d - c}{b - a}$$

(i.e. probability is *proportional* to length of interval).

We can determine the cumulative distribution function F_X and the density f_X for X as follows:

First notice that:

1. For $x < a$, $F_X(x) = \mathbb{P}(X \leqslant x) = 0$ (since X cannot be smaller than a).
2. For $x \geqslant b$, $F_X(x) = \mathbb{P}(X \leqslant x) = 1$ (since for any $x \geqslant b$, X is definitely no greater than x).

Then for $a \leqslant x < b$,

$$F_X(x) = \mathbb{P}(X \leqslant x) = \mathbb{P}(a \leqslant X \leqslant x) \quad = \quad \frac{x - a}{b - a}.$$

So for $X \sim \mathrm{U}(a,b)$, the cumulative distribution function is

$$\boxed{F_X(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leqslant x < b \\ 1 & x \geqslant b \end{cases}}$$

To find the probability density function for $X \sim \mathrm{U}(a,b)$, differentiate the cumulative distribution function:

$$\boxed{f_X(x) = \frac{dF_X}{dx} = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & \text{otherwise} \end{cases}}$$

Notice that the density f_X is constant on (a,b).

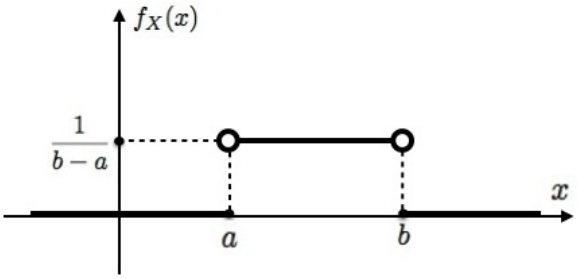


Figure 3.7: Sketch of $f_X(x)$

Properties of Uniform random variables
For $X \sim \mathrm{U}(a,b)$:

$$\mathrm{Im}X = [a,b], \qquad F_X(x) = \begin{cases} 0, & x < a, \\ \frac{x-a}{b-a}, & a \leqslant x < b, \\ 1, & b \leqslant x, \end{cases} \qquad f_X(x) = \begin{cases} \frac{1}{b-a}, & a < x < b, \\ 0, & \text{otherwise.} \end{cases}$$

$$\mathbb{E}(X) = \frac{a+b}{2} \qquad \mathbb{V}\mathrm{ar}(X) = \frac{(b-a)^2}{12}$$

Suppose $X \sim \mathrm{U}(a,b)$ and $Y = cX + d$ where $c > 0$. Then $Y \sim \mathrm{U}(ca+d, cb+d)$.
That is, a linear function of a uniform variable is *also* uniform.

Proof: With $X \sim \mathrm{U}(a,b)$ and $Y = cX + d$ where $c > 0$, since $\mathrm{Im}X = [a,b]$ then $\mathrm{Im}Y = [ca+d, cb+d]$. Hence the cumulative distribution function of Y is given by $F_Y(y) = 0$ for

$y < ca + d$, and $F_Y(y) = 1$ for $y \geqslant cb + d$. For $ca + d \leqslant y < cb + d$,

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \leqslant y) \\ &= \mathbb{P}(cX + d \leqslant y) \\ &= \mathbb{P}\left(X \leqslant \frac{y-d}{c}\right) \\ &= F_X\left(\frac{y-d}{c}\right) \\ &= \frac{\frac{y-d}{c} - a}{b-a} \\ &= \frac{y-d-ca}{cb-ca} \\ &= \frac{y-(ca+d)}{(cb+d)-(ca+d)} \end{aligned}$$

So Y has cumulative distribution function

$$F_Y(y) = \begin{cases} 0 & y < ca + d \\ \frac{y-(ca+d)}{(cb+d)-(ca+d)} & ca + d \leqslant y < cb + d \\ 1 & y \geqslant cb + d \end{cases}$$

which is the cumulative distribution function of the $\mathrm{U}(ca + d, cb + d)$ distribution.

3.3.2 Exponential distribution

Exponential random variables arise in many areas of statistics and actuarial science; in particular, they can be used to model random waiting times. We say that X has an exponential distribution with parameter $\lambda > 0$, and write $X \sim \mathrm{Exp}(\lambda)$, if and only if its density is given by

$$\boxed{f_X(x) = \begin{cases} 0 & x < 0 \\ \lambda e^{-\lambda x} & x \geqslant 0 \end{cases}}$$

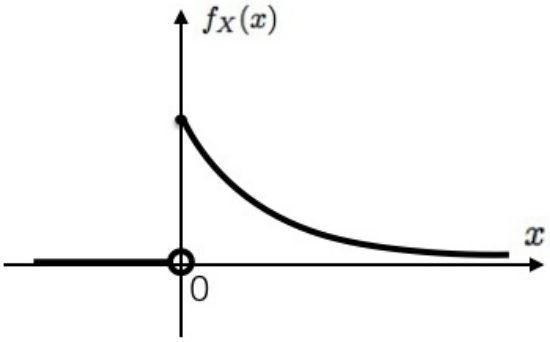


Figure 3.8: Sketch of the density $f_X(x)$

We note that $\mathrm{Im}(X) = [0, \infty)$ and that the cumulative distribution function F_X for $X \sim \mathrm{Exp}(\lambda)$ is given by

$$\boxed{F_X(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-\lambda x} & x \geqslant 0 \end{cases}}$$

An Important Property of Exponential Random Variables
Suppose $X \sim \mathrm{Exp}(\lambda)$ and $c > 0$. Let $Y = cX$, then $Y \sim \mathrm{Exp}(\lambda/c)$. That is, Y is *also* an exponential random variable.

Proof: Since $X \sim \text{Exp}(\lambda)$ we have $\text{Im}X = [0, \infty)$, so with $c > 0$ and $Y = cX$, then $\text{Im}Y = [0, \infty)$. So for $y < 0$ we have $F_Y(y) = 0$. To find $F_Y(y)$ for $y \geqslant 0$,

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \leqslant y) \\ &= \mathbb{P}(cX \leqslant y) \\ &= \mathbb{P}(X \leqslant y/c) \\ &= F_X\left(y/c\right) \\ &= 1 - e^{-\lambda(y/c)} \end{aligned}$$

So the distribution function of Y is

$$F_Y(y) = \begin{cases} 0 & y < 0 \\ 1 - e^{-(\lambda/c)y} & y \geq 0 \end{cases}$$

This is the distribution function of the $\text{Exp}(\lambda/c)$ distribution, so $Y \sim \text{Exp}(\lambda/c)$

Remark 3.2. Exponential random variables arise as models for the length of time (starting from now) until a 'completely random' event occurs. For example, exponential random variables are often used to model:

- the time until an earthquake occurs
- the time until next telephone call is received
- the time until a particle is emitted from a radioactive substance

Example 3.4. The lifetime X (in thousands of hours) of a particular electrical component may be modelled as an exponential random variable with mean 2000 hours. Calculate the probability that the component's lifetime is less than 1000 hours.

Answer: In units of thousands of hours, we are told that $\mathbb{E}(X) = 2$. For $X \sim \text{Exp}(\lambda)$ we know that $\mathbb{E}(X) = 1/\lambda$, so in this example we have $\lambda = 1/2$, and X has distribution function

$$F_X(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-(1/2)x} & x \geq 0 \end{cases}$$

The required probability is

$$\begin{aligned} \mathbb{P}(X < 1) &= \mathbb{P}(X \leqslant 1) \qquad \text{since } X \text{ is a } \textit{continuous} \text{ random variable} \\ &= F_X(1) \\ &= 1 - e^{-1/2} \approx 0.393 \end{aligned}$$

3.3.3 The Gamma distribution

Exponential random variables (see Section 3.3.2) are a special case of a more general class of random variables called gamma random variables. A random variable X has a gamma distribution with parameters $\alpha > 0$ and $\lambda > 0$, written $X \sim \text{Gamma}\,(\alpha, \lambda)$, if and only if its density is given by

$$\boxed{f_X(x) = \begin{cases} 0 & x < 0 \\ \frac{\lambda^\alpha}{\Gamma(\alpha)}\, x^{\alpha-1} e^{-\lambda x} & x \geqslant 0 \end{cases}}$$

where the Gamma function $\Gamma(\alpha)$ is defined by

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} \mathrm{d}x \text{ for } \alpha > 0$$

Exercise: Show that the function $f_X(x)$ given above is a valid probability density function.

The Gamma function is related to the factorial function as follows. Suppose that $\alpha > 1$. Then applying integration by parts in the definition of the Gamma function we can show that

$$\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1)$$

Now, since $\Gamma(1) = 1$ (check this!), it is straightforward to deduce (by induction) that if α is a positive integer then

$$\Gamma(\alpha) = (\alpha - 1)!$$

The Gamma function can be considered to be an extension of the factorial function to non-integer values.

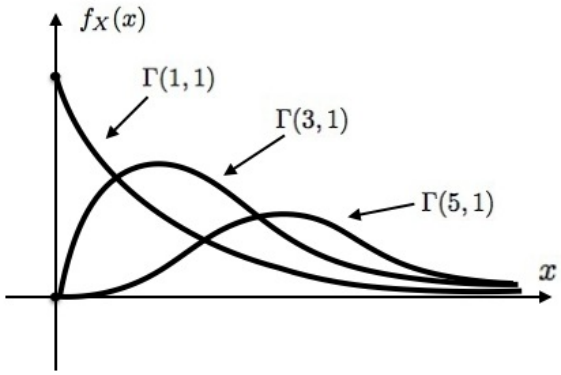
The relationship $\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1)$ is very useful in many situations as it enables to do many important calculations without explicitly evaluating $\Gamma(\alpha)$. In particular, it follows that, for $\alpha > 0, k \in \mathbb{Z}, k > 0$,

$$\frac{\Gamma(\alpha + k)}{\Gamma(\alpha)} = \alpha(\alpha + 1) \times \cdots \times \ldots (\alpha + k - 1)$$

Remarks

1. Gamma random variables provide a family of models for positive quantities such as survival times, etc

2. The parameter α is called the shape parameter and the parameter λ is called the reciprocal scale parameter or the rate parameter.



3. The exponential distribution is a special case of the gamma distribution. In particular, for any $\lambda > 0$, if $X \sim \text{Exp}(\lambda)$, then X has a Gamma $(1, \lambda)$ distribution.